

BIOINFORMATICS

WEEK 1 – DAY 3

LINUX is an open-source operating system (OS) that directly manages a computer's hardware and resources, acting as the fundamental software layer between your applications and the physical machine.

Why Learn Linux?

Because:

- Most bioinformatics software is built for Linux.
- High-performance computing (HPC) clusters usually run Linux.
- Sequencing data is often analyzed on Linux servers.
- Tools like BWA, Bowtie, HISAT2, SAMtools, and GATK are commonly used in Linux environments.

If Python is your language, Linux is your workspace.

DO YOU NEED TO REPLACE WINDOWS?

No. The easiest option is to use WSL (Windows Subsystem for Linux). It lets you run Ubuntu inside Windows. You'll have both Windows and Linux on the same laptop.

INSTALL WSL

Open PowerShell as Administrator.

Run: **wsl - install**

Restart your computer if prompted.

After restarting, Ubuntu will launch and ask you to create Username & Password

UNDERSTANDING THE LINUX FILE SYSTEM

- Windows uses drives: C:\ & D:\
- Linux uses one unified filesystem. The root is: /

The shortcut: ~ [means "my home directory."]

GETTING STARTED WITH LINUX AND BASH

Installing and Updating Software

1. In Ubuntu, for example, first update the list of software available by running `sudo apt update`. Then, you can directly get software by using the `sudo apt-get install` command followed by the name of the program you wish to install:

Bash

```
sudo apt-get install <app_name>
```

2. To update programs that have already been installed, you can run:

Bash

```
sudo apt update && sudo apt upgrade
```

WORKING WITH FILES AND DIRECTORIES

1. To view the path of the directory you are currently in, use the `pwd` command:

Bash

```
pwd
```

2. To create a new directory, use the `mkdir` command followed by the name of the directory you want to create:

Bash

```
mkdir hello_world
```

3. To change directories, use the `cd` command followed by the name of the directory you want to navigate to:

Bash

```
cd hello_world
```

4. To see the contents within the directory you are currently in, type `ls` into the command line:

Bash

```
ls
```

5. By default, the `ls` command will print the name of all the files and directories only. To get additional information such as the last time a file was modified or file permissions, use the flag `-l`:

Bash

```
ls -l
```

6. You can create a new file via the `touch` command followed by the name of the file you would like to create:

Bash

```
touch hello_world.txt
```

7. You can edit files using any downloaded graphical text-editor or the VS Code Remote – WSL extension.

To edit your file with your preferred method of editing, simply run the program name followed by the name of the file you'd like to edit:

Bash

```
code hello_world.txt
```

Bash

```
notepad.exe hello_world.txt
```

8. To see the contents of a file in the command line, use the `cat` command followed by the file you'd like to read:

Bash

```
cat hello_world.txt
```

9. To Write Into a File

Quickly write text: `echo "Hello Bioinformatics" > hello.txt`

```
cat hello.txt
```

Output: `Hello Bioinformatics`

Append more text: `echo "Python is fun." >> hello.txt`

```
cat hello.txt
```

Output: `Hello Bioinformatics`

```
Python is fun.
```

Notice the difference:

- > creates or overwrites a file.
- >> adds to the end without deleting existing content.

10. To Copy Files

```
cp hello.txt backup.txt  
ls
```

You'll have both files.

11. To Rename or Move

```
mv backup.txt old_notes.txt
```

The same command can move a file into another folder.

12. To Delete Files

```
rm old_notes.txt
```

Be careful—Linux doesn't have a Recycle Bin for rm.

13. To Search for Text

Suppose dna.fasta contains: `ATGCGATCGATCGATCG`

Find lines containing ATG: `grep ATG dna.fasta`

14. To View Part of a File

First few lines: `head dna.fasta`

Last few lines: `tail dna.fasta`

15. To Find Files

`find . -name "*.fasta"`

This searches the current directory (.) and all subdirectories.

Command	Purpose
pwd	Show current location
ls	List files
cd	Change directory
mkdir	Create folders
touch	Create files
cat	Display file contents
echo	Write text to files
cp	Copy files
mv	Move or rename files
rm	Delete files
head	Show first lines
tail	Show last lines
grep	Search inside files
find	Locate files